

UPSC Civil Services (Main) Simulation GS-III

DETAILED SOLUTIONS / ANSWER KEY | Time: 3 Hours | Maximum Marks: 250

Q1. Distinguish GDP, GVA, GNI and NDP.

[10 marks | 150 words]

Model answer (143 words; practice model, not official): GDP measures final production within domestic territory. GVA measures producers' value added: output minus intermediate consumption. GDP at market prices equals GVA at basic prices plus taxes on products minus subsidies on products. GNI, corresponding to legacy GNP, adds net primary income from abroad to GDP. NDP subtracts consumption of fixed capital from GDP and shows domestic production net of used-up fixed assets. The distinctions matter analytically. A change in net product taxes can separate GDP growth from GVA growth; negative net primary income from abroad can make GNI lower than GDP; and gross output can overstate the amount remaining after capital maintenance. Thus an answer should identify four axes before comparison: domestic or national, gross or net, basic or market prices, and nominal or real. The correct aggregate follows the question's boundary, not a hierarchy in which one measure is always superior.

Q2. Explain why digital land records do not automatically create conclusive title.

[10 marks | 150 words]

Model answer (117 words; practice model, not official): Digital records improve access, map consistency, mutation tracking and registration links. DILRMP 3.0 guidelines cover 2026-2031 and integrate textual records, cadastral maps and services. ULPIN identifies a georeferenced parcel, while SVAMITVA supports drone mapping and property cards in rural abadi areas. These tools strengthen evidence, but title is a legal conclusion. Records may omit tenants, women, customary users or community rights; boundaries may be disputed; and State law determines a card's effect. Conclusive title additionally requires adjudication, notice, objections, continuous updating and rules allocating loss when the register is wrong. Digitisation should therefore include participatory survey, accessible correction and independent dispute resolution. Technology reduces uncertainty, but treating a mapped record as infeasible can formalise old errors.

Q3. Distinguish public-private partnership from privatisation.

[10 marks | 150 words]

Model answer (126 words; practice model, not official): A PPP is a fixed-term service contract allocating investment, management, performance and risks between a public authority and private entity. Privatisation transfers ownership. The distinction affects accountability. Under BOT or DBFOT, the concessionaire may finance, build and operate, yet the asset returns to government. Under OMT or TOT, only operating and revenue rights over an existing asset are transferred for a term. PPP is therefore neither free infrastructure nor withdrawal of the State. Government retains policy, regulation, land, affordability and service-continuity duties. Users or taxpayers still fund the service. The correct test is contractual: identify ownership, functions, payment source, risk allocation, duration, standards and handback. A long concession may resemble private control, but it remains different from permanent sale when public ownership and service obligations survive.

Q4. Distinguish disinvestment, strategic sale, privatisation and asset monetisation.

[10 marks | 150 words]

Model answer (123 words; practice model, not official): Disinvestment is sale of part or all of government's equity. A minority IPO or Offer for Sale reduces shareholding but leaves management control with government. Strategic disinvestment adds management-control transfer to a substantial or entire sale. If control passes to a private buyer, it is privatisation. Air India's January 2022 transfer to Talace illustrates this route. Transfer to another CPSE can be strategic disinvestment without privatisation. Asset monetisation transfers operating or revenue rights, often temporarily, while public ownership may remain. It is not necessarily an equity sale. ETF and buyback are minority or capital-management routes. Trace the object, duration, buyer, equity and control. Valuation and accountability therefore differ across routes. Receipts are one-off capital receipts, not proof of recurring fiscal or enterprise improvement.

Q5. Distinguish the Consolidated Fund, Public Account and Contingency Fund of India.

[10 marks | 150 words]

Model answer (138 words; practice model, not official): Article 266 places Union revenues, loans raised and loan recoveries in the Consolidated Fund of India. Government expenditure is ordinarily incurred from this Fund, and withdrawal requires parliamentary authorisation through appropriation. Charged expenditure is not voted, but it is still discussed, disclosed and appropriated. Article 266(2) creates the Public Account for money held by government in trust, including provident funds, small savings and specified

deposits. Such money is repayable; routine repayment does not require parliamentary appropriation, though transfers into the account from the Consolidated Fund need authority. Article 267 permits the Contingency Fund as an imprest at the President's disposal for urgent unforeseen expenditure pending legislative approval. After ex-post authorisation, an equivalent amount is drawn from the Consolidated Fund to recoup it. Thus the three differ by ownership, purpose and withdrawal rule. None permits uncontrolled executive expenditure.

Q6. Explain why energy flow is unidirectional while matter cycles in an ecosystem.

[10 marks | 150 words]

Model answer (90 words; practice model, not official): 1. Producers introduce usable chemical energy through photosynthesis or chemosynthesis; green plants are not the only producers, and the evidence does not justify treating consumers or decomposers as an energy entry point. 2. Detritivores physically fragment dead organic material, whereas microbial decomposers such as bacteria and fungi chemically break it down and release nutrients; the two roles cooperate but are not synonyms. 3. Energy passes one way through trophic transfers and dissipates as heat, whereas chemical matter returns through decomposition and uptake; energy is not recycled merely because nutrients are.

Q7. Differentiate PSLV, GSLV, LVM3 and SSLV without reducing the comparison to payload size.

[10 marks | 150 words]

Model answer (144 words; practice model, not official): 1. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. 2. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. 3. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. 4. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. 5. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Q8. Distinguish internal security from external security in India's federal setting.

[10 marks | 150 words]

Model answer (141 words; practice model, not official): 1. Internal security concerns peace, law and order, rule of law and sovereignty within India's territory; external security addresses foreign aggression, but neighbourhood linkages make the two domains inter-related. 2. The Ministry of Home Affairs is the Union ministry responsible for internal security, while the Ministry of Defence owns external defence; central assistance does not erase State policing competence. 3. Public order is State List Entry 1 and police is State List Entry 2, making the State the primary day-to-day authority for internal order and investigation. 4. Union List Entry 2A permits deployment of an armed force of the Union in a State in aid of the civil power; assistance supplements rather than replaces the State authority. 5. Article 355 places a duty on the Union to protect every State against external aggression and internal disturbance and to ensure constitutional government.

Q9. Differentiate NDRF-the-force from disaster-response and mitigation funds.

[10 marks | 150 words]

Model answer (101 words; practice model, not official): 1. The National Disaster Response Force is a specialist response force under the Act; official NDRF material states that it has expanded to 16 battalions. 2. The National Disaster Response Fund under section 46 finances eligible response and relief needs and is institutionally distinct from the identically abbreviated response force. 3. Section 44A enables a State Government to constitute a specialist State Disaster Response Force; this force must not be confused with the SDRF fund. 4. Sections 46, 47 and 48 contemplate national, state and district response and mitigation funds; statutory provision does not establish that every district fund is operational.

Q10. Compare CPI, WPI and the GDP deflator as inflation measures.

[10 marks | 150 words]

Model answer (134 words; practice model, not official): CPI measures retail prices of a representative household-consumption basket. It includes goods, services and imported consumption; headline CPI is India's policy anchor. MoSPI introduced the current 2024=100 series on 12 February 2026 with HCES 2023-24 weights. WPI, compiled by the Office of Economic Adviser, DPIIT, tracks wholesale/basic prices of goods and excludes services. Its 2022-23-base series was released for May 2026. It signals commodity and pipeline cost pressure but is not a household cost-of-living index. The GDP deflator equals nominal GDP divided by real GDP multiplied by 100. It covers domestically produced final goods and services, excludes imports directly and has a changing production mix. It is broad but revised with national accounts and less timely monthly. Thus CPI suits consumer welfare and targeting, WPI pipeline analysis, and the deflator domestic-output prices.

Q11. Analyse India's regulatory architecture for money and capital markets. Why is a functional rather than slogan-based jurisdiction map necessary?

[15 marks | 250 words]

Model answer (221 words; practice model, not official): India's market architecture divides responsibility by instrument, issuer, maturity, venue and activity. The Reserve Bank of India leads money markets, Government securities, foreign exchange and payment systems under its statutory mandates. It issues directions for call, notice and term money, Certificates of Deposit, Commercial Paper, repo and the Negotiated Dealing System-Order Matching platform. It also manages government borrowing and supervises regulated participants. The Securities and Exchange Board of India leads public issuance and trading in equity and corporate securities, recognised stock exchanges, clearing corporations, depositories and listed-market conduct. The Companies Act and government borrowing framework add issuer-side rules. Exchanges organise trading, clearing corporations manage counterparty and settlement risk, and depositories maintain electronic ownership records; none should be described as the other. Functional mapping is necessary because simple formulas fail. Debt equals RBI is false: listed corporate bonds fall substantially within SEBI's perimeter, while short-term Commercial Paper is an RBI-regulated money-market instrument. An electronic platform may facilitate an over-the-counter market, and a single transaction can involve a regulator, venue, central counterparty and depository. Specialisation is the advantage; boundary gaps, inconsistent disclosure and regulatory arbitrage are the risks. Coordination should use activity-based definitions, common identifiers, interoperable data and clear crisis protocols. The aim is complete coverage: issuance integrity, prudential resilience, transparent trading, final settlement, custody and investor remedy each need an accountable owner.

Q12. Discuss reforms needed to make TPDS effective, transparent and exclusion-safe.

[15 marks | 250 words]

Model answer (197 words; practice model, not official): TPDS converts NFSA allocation into household receipt through State depots and fair-price shops. Reform must therefore address lists, logistics, transactions and remedies together. First, beneficiary databases need regular, appealable updating so deaths and duplicates are removed without excluding migrants, elderly people, women or newly poor households. One Nation One Ration Card should be supported by interoperable stocks, beneficiary awareness and split-family portability. Second, end-to-end computerisation, GPS or stock tracking and ePoS records can expose diversion, but biometric or network failure must trigger alternate authentication rather than denial. Third, allocation, offtake, fair-price-shop stock and household receipt should be disclosed in accessible formats. Social audits, vigilance committees, District Grievance Redressal Officers and State Food Commissions need time-bound powers and public compliance reporting. Dealer margins and shop viability also affect service quality. Fourth, food quality, timely movement and doorstep delivery require stronger warehouse and transport accountability. Nutrition-sensitive additions must respect local diets and reliable supply. Cash transfer can reduce handling where banking and competitive food markets are dependable, but inflation indexation, payment delay and intra-household control create risks. Hence reform should preserve an in-kind fallback and judge success by actual receipt and reduced hunger, not merely digitisation or deletion counts.

Q13. Analyse the economics of integrating variable renewable energy into India's power system.

[15 marks | 250 words]

Model answer (175 words; practice model, not official): Variable renewable energy has low fuel and short-run marginal cost, but output depends on weather and time. The problem shifts from procuring cheap plant-level energy to delivering reliable system service. At 31 December 2025, non-fossil sources formed 51.93% of installed capacity, while their April-December 2025 generation share was 30.41%; the different periods notwithstanding, this shows why nameplate capacity is not firm supply. Integration requires advance transmission, accurate forecasting, larger balancing areas, flexible hydro and thermal operation, storage, demand response, reserves and short-term markets. MNRE's GEC-II design aims at 10,750 circuit-km and 27,500 MVA to integrate about 20 GW. Economic Survey 2025-26 cites storage need of 336 GWh by 2029-30 and 411 GWh by 2031-32. Costs include curtailment, networks, balancing, land, minerals and recycling; benefits include avoided fuel, pollution and imports. Policy should procure portfolios and services rather than isolated megawatts: generation plus evacuation, duration-specific storage, time-of-day demand and resource adequacy. The verdict is pro-integration: success is lower unserved energy and total system cost, not the lowest bid or largest capacity addition.

Q14. Critically evaluate Production Linked Incentive schemes through additionality and domestic value addition.

[15 marks | 250 words]

Model answer (200 words; practice model, not official): PLI links public support to eligible incremental production or sales, seeking scale, learning, investment and strategic resilience. Its strength is ex-post verification: a factory announcement alone earns nothing. Named evidence is the fourteen-sector umbrella with about Rs 1.97 lakh crore announced outlay. DPIIT's 10 December 2025 review, using data through June 2025, reported investment, incremental sales, exports and employment. These show implementation scale, not causal success. Evaluation must separate outlay, approval, investment, production, claim and disbursement. Economic additionality asks whether activity would have occurred without support. Domestic value addition subtracts imported inputs; gross exports may remain assembly-heavy. Jobs require payroll, wage, skill and displacement checks. Risks include favourable base years, deadweight subsidy, concentration among incumbents, exclusion of smaller suppliers, import-heavy output and pressure for

extension. WTO exposure depends on the legal trigger, especially export or local-content contingency. Sector comparisons should normalise investment cycles, price changes and market growth. Publishing rejected claims, supplier outcomes and beneficiary concentration would improve accountability and reveal whether verification is strict. PLI should therefore retain sector-specific milestones, transparent beneficiary selection, competition review, supplier-development plans, independent counterfactual evaluation and a credible sunset. It is useful only if temporary fiscal support creates post-incentive productivity and technological depth.

Q15. Analyse the constitutional balance between the GST Council's harmonising role and legislative autonomy after Mohit Minerals.

[15 marks | 250 words]

Model answer (198 words; practice model, not official): The GST Constitution combines shared coordination with legislative autonomy. Article 246A gives Parliament and State legislatures simultaneous GST power, while Parliament has exclusive power over inter-State supplies. Article 279A creates the GST Council to recommend rates, exemptions, model laws, thresholds and place-of-supply principles. Its weighted voting gives one-third to the Union and two-thirds collectively to States, with a three-fourths decision threshold. In *Union of India v Mohit Minerals Pvt Ltd*, decided on 19 May 2022, the Supreme Court held that Council recommendations have persuasive value but are not binding on Union and State legislatures. Binding recommendations would dilute the simultaneous authority conferred by Article 246A. Yet the Court recognised that GST depends on dialogue and cooperative federalism. This balance supports harmonisation without creating an extra-constitutional legislature. Common recommendations reduce trade barriers, classification divergence and compliance costs; retained legislative accountability preserves democratic choice. However, persistent divergence could fragment credit and settlement even when constitutionally permissible. Thus, non-binding does not mean irrelevant. Council consensus needs transparent data, reasoned recommendations, valid legislation and clear notifications. Judicial review and legislative autonomy remain safeguards, while political cooperation makes the system work. GST is negotiated harmonisation, neither Union command nor unrestricted State tax competition.

Q16. Explain the CITES science-to-permit chain and its institutional division of labour.

[15 marks | 250 words]

Model answer (138 words; practice model, not official): 1. The Scientific Authority's non-detriment finding is the science gate for relevant exports and must remain distinct from the Management Authority's decision to issue a permit or certificate. 2. Permit scrutiny also requires the applicable authority to be satisfied about lawful acquisition where the Convention and domestic implementation require it; an NDF alone is not the whole permit test. 3. A national Management Authority administers permits and certificates; it does not replace the Scientific Authority's conservation advice or customs' border-verification role. 4. A national Scientific Authority advises on whether trade is detrimental and on related scientific questions; it does not issue the trade document merely because it supplied advice. 5. Effective control depends on the chain from scientific finding and permit issuance to species identification, customs inspection, intelligence and cross-border cooperation; listing alone does not verify a shipment.

Q17. Assess constellation health and receiver adoption as constraints on NavIC.

[15 marks | 250 words]

Model answer (163 words; practice model, not official): 1. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. 2. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. 3. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. 4. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. 5. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. 6. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome.

Q18. Analyse the UAPA and NIA Act amendments in the prevailing security environment.

[15 marks | 250 words]

Model answer (221 words; practice model, not official): 1. The NIA Act, 2008 creates a central investigating agency for scheduled offences with concurrent jurisdiction; NIA does not replace the local police's first response, evidence preservation or ordinary policing role. 2. The proposed NCTC faced objection because independent search, arrest and investigation powers under an intelligence body could bypass State policing competence; the coordination need did not automatically justify centralisation. 3. Local police ordinarily reach an attack first and secure the scene, witnesses, evidence and public order, making State capacity structurally irreplaceable even in nationally investigated cases. 4. The UAPA (Amendment) Act, 2019 permits the Central Government to designate an individual as a terrorist in the Fourth Schedule through executive notification and a statutory review or denotification route. 5. Individual designation, arrest, charge-sheet, trial and conviction are five distinct legal and evidentiary events; designation is not a judicial finding of guilt. 6. The NIA (Amendment) Act, 2019 extends jurisdiction to specified offences outside India affecting Indian citizens or interests, expands scheduled coverage and permits Sessions Courts to be designated Special Courts. 7. UAPA

section 43D(5) creates a stringent prima-facie-true bail threshold; pre-trial liberty, prosecution and conviction must be analysed separately. 8. TADA operated from 1985 to 1995 and POTA from 2002 to 2004 before lapse or repeal amid misuse concerns, illustrating the recurring necessity-accountability tension in special anti-terror law.

Q19. Analyse social vulnerability through gender, age, disability and displacement.

[15 marks | 250 words]

Model answer (139 words; practice model, not official): 1. Poverty, gender, age, disability, health, livelihood, housing, documentation and displacement can produce differentiated impacts from the same hazard. 2. Risk can compound when vulnerabilities intersect, so a single label such as vulnerable sections is analytically insufficient. 3. Gender-responsive DRR requires participation in planning and attention to privacy, sanitation, safety, health, livelihoods, unpaid care and protection from violence. 4. Children need age-appropriate warnings, family tracing, safe learning continuity, nutrition, health care and protection from separation, abuse and trafficking. 5. Older persons may require medication continuity, mobility assistance, accessible transport, caregiver support and inclusion in household and shelter registers. 6. Disability-inclusive DRR requires accessible information, assisted evacuation, barrier-free shelters, continuity of support and participation in decisions. 7. Migrants and displaced people may face language, documentation, rental, livelihood and exclusion barriers; disaster displacement does not automatically create 1951 Refugee Convention status.

Q20. Compare forwards, futures, options and swaps as instruments of risk transfer.

[15 marks | 250 words]

Model answer (226 words; practice model, not official): Forwards and futures create symmetric obligations to transact later at an agreed price. A forward is customised and usually over the counter, providing a close hedge match but leaving bilateral valuation, documentation and counterparty exposure. A future is standardised, exchange-traded and centrally cleared. Daily mark-to-market and margin reduce accumulated counterparty risk but create immediate liquidity demands. Options create asymmetric rights. A call buyer may buy and a put buyer may sell at the strike; either can allow the option to expire. The buyer pays a premium and, for a simple long option held to expiry, limits contractual loss to that premium. The writer receives premium but bears the contingent obligation and may face large margin calls. Strike, premium, time and volatility shape payoff. Swaps exchange cash-flow streams, such as fixed for floating interest or specified currency-linked flows. The reference notional may calculate payments without being exchanged. Swaps can transform balance-sheet exposure without refinancing the original liability, but basis and counterparty risks remain. All four can hedge or speculate. Exchange standardisation supports liquidity and transparent risk management; OTC customisation can reduce basis risk. SEBI governs exchange-traded securities derivatives, while RBI governs specified OTC rate and currency derivatives. Sound regulation therefore combines reporting, collateral, position controls, central clearing where appropriate and suitability. The test is not product complexity alone but whether residual exposure and liquidity demands are understood.

END OF DETAILED SOLUTIONS